

Work, Energy, Power.

VI

Work in 1-D

Work done on a body is defined as the product of force and displacement (in the direction of the force).

$$W = Fx \quad \text{Units} \quad \text{Joule } J = Nm.$$

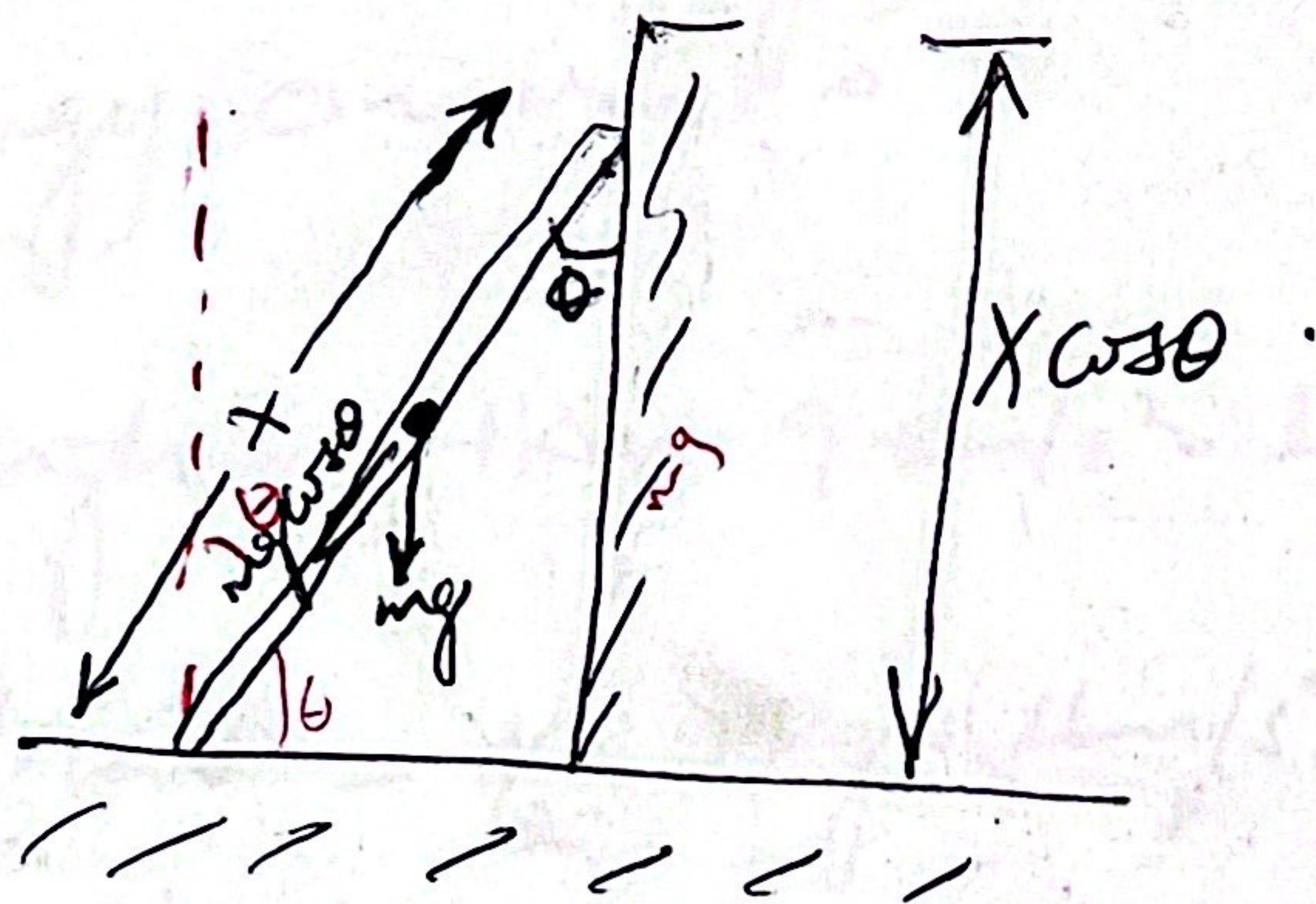
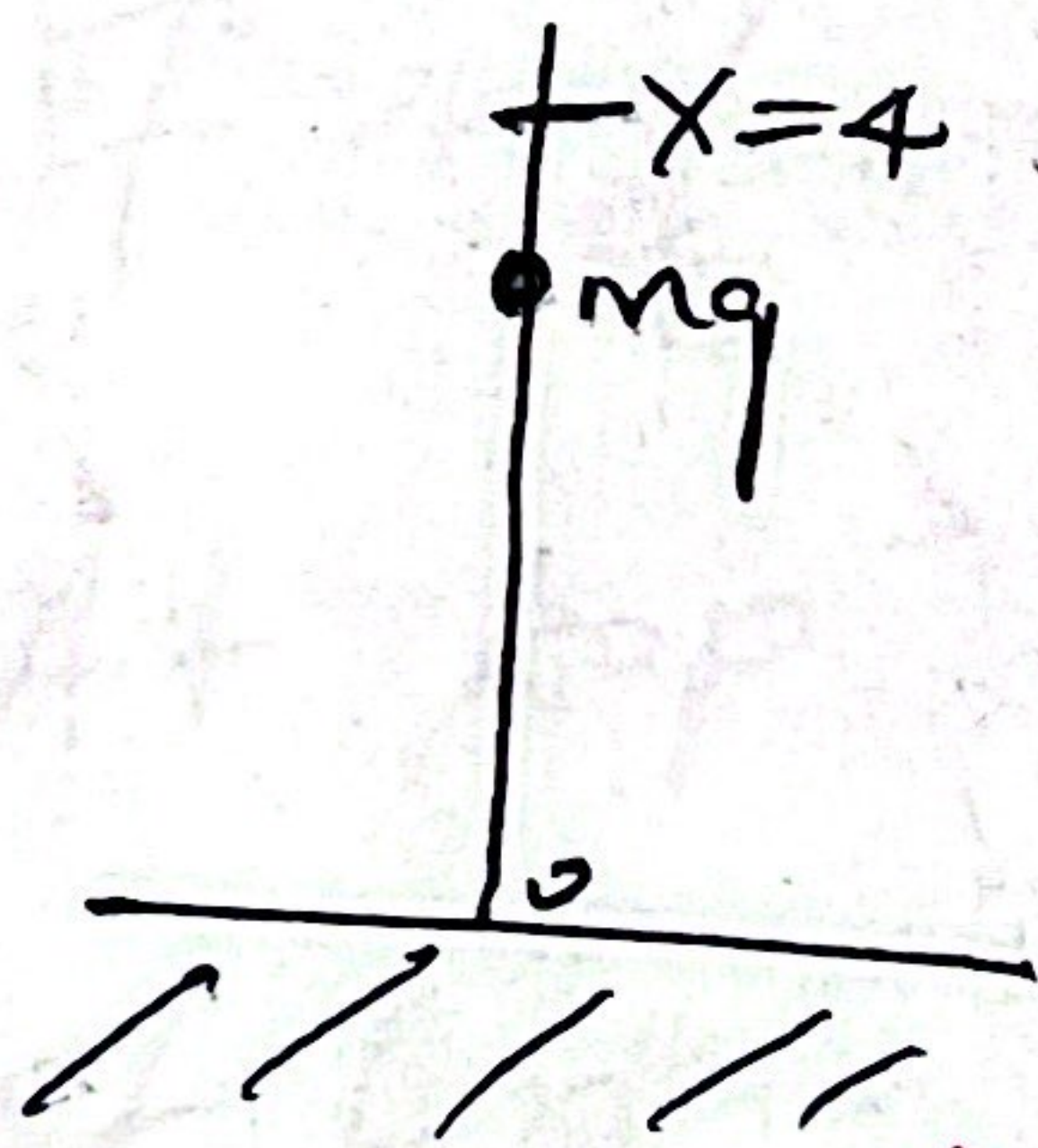
When force = 0 no work is done, similarly when $x = 0$ no work is done.

E.g.

Compute the work done as a man of weight 750 N climbs a ladder through a height of 4 m.

Sol.

$$W = Fx = (750 N)(4 m) = 3000 J.$$



$$W = Fx \cos \theta \quad (x \cos \theta = \text{displacement involved})$$

Work done on a body can set it into motion, or can place it in a position such that future motion is possible. In either case, the object is said to possess the property called energy.

Energy : Forms & sources

Energy is the capacity to perform work.

Potential Energy — is the energy associated with position

$$PE = F \times h = mgh.$$

Kinetic Energy — the energy possessed by an object by virtue of its motion

$$KE = \frac{1}{2}mv^2.$$

Example

A 1kg block slides down a rough inclined plane whose height is 1m. At the bottom the block has a velocity of 4m/s. Is energy conserved?

At top $PE = mgh = 1\text{kg} \cdot 9.8\text{m/s}^2 \cdot 1\text{m} = 9.8\text{J}$

At bottom $KE = \frac{1}{2}mv^2 = \frac{1}{2} \cdot 1\text{kg} \cdot (4\text{m/s})^2 = 8\text{J}$

∴ not conserved but appears as thermal energy.

Energy = capacity to perform work.

Forms and sources of Energy:

Energy ~~that~~ stored that can be released to yield energy of motion is classified as potential energy. E.g.

- metal spring
- storage battery
- water reservoir
- chemicals
- nuclear fuel

P.E is defined as the measure of capacity of a system to do work. It's the energy associated with position.

$$E_p = Fx \quad \text{Nm} \quad \text{E.g. gravitational potential. E.}$$

$mgh = PE_{\text{grav}}$

The work done in setting a body in motion is given the special name Kinetic energy

$KE = \frac{1}{2}mv^2$

KE_{rot} — Spin with c of mass stationary

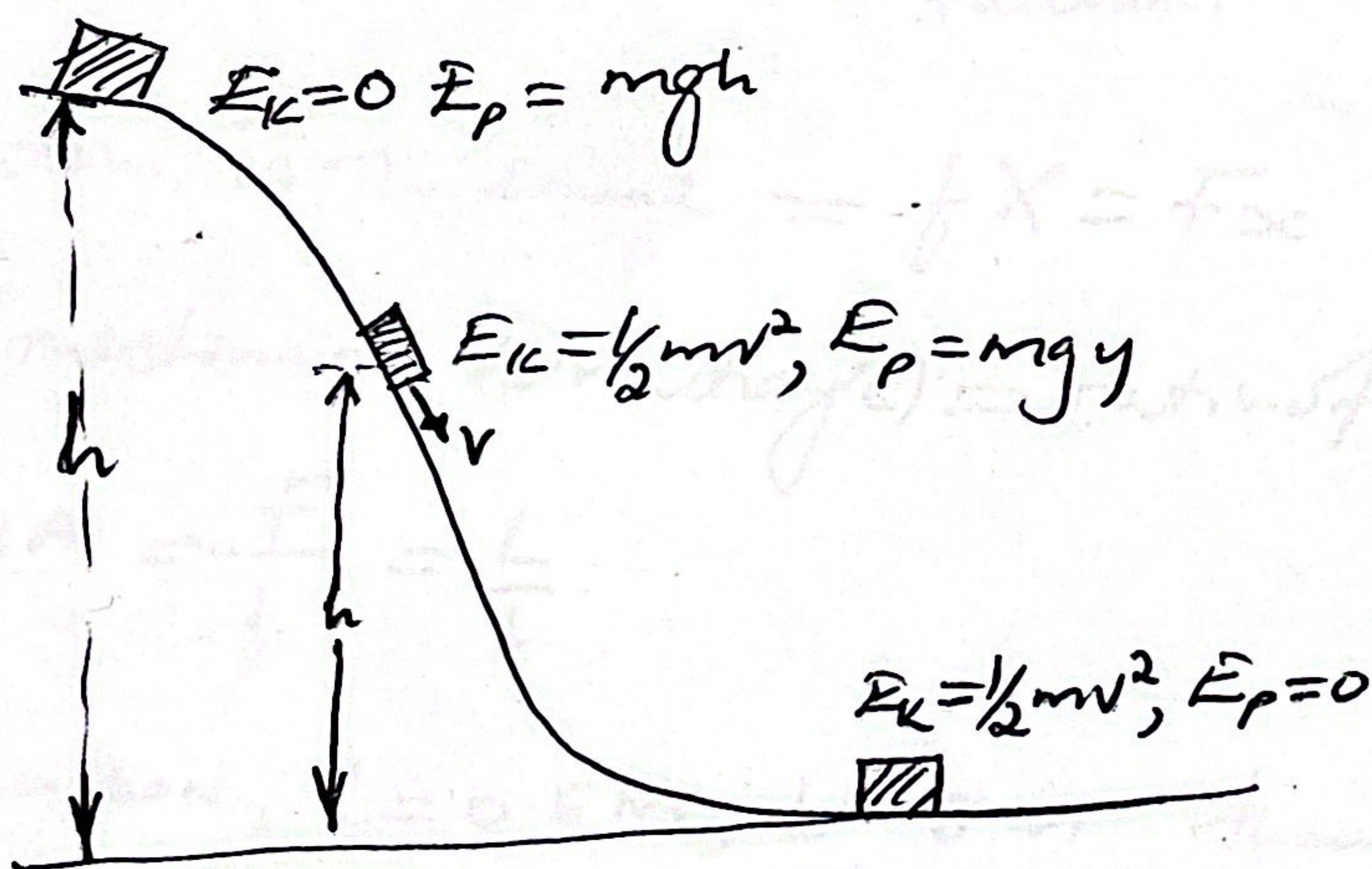
KE_{tran} — Centre of mass moving

E.g. $m = 2\text{kg}$ initially at rest

* K.E. can be translational or rotational. In translational, the C.O.F. mass moves. In rotational, C.O.F. mass does not move but the object spins.

Conservation of energy — ~~either~~ the loss of energy anywhere is always compensated by an equal gain of energy somewhere else.

Eg. a mass ~~not~~ falling thro' a height ~~height~~ acquires K.E.



Gravitational potential.

Near the surface of the earth, $PE_g \approx mgh$, but at some height r from the surface,

$$PE_g = -\frac{GMm}{r} \quad \text{or} \quad PE_g = \frac{GMm}{r_1} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad r_2 > r_1$$

Forms and Sources of Energy.

The main source of energy are the sun and other stars. Other sources are: Geothermal, coal, oil, biogas, nuclear reactions, hydroelectric and wind.

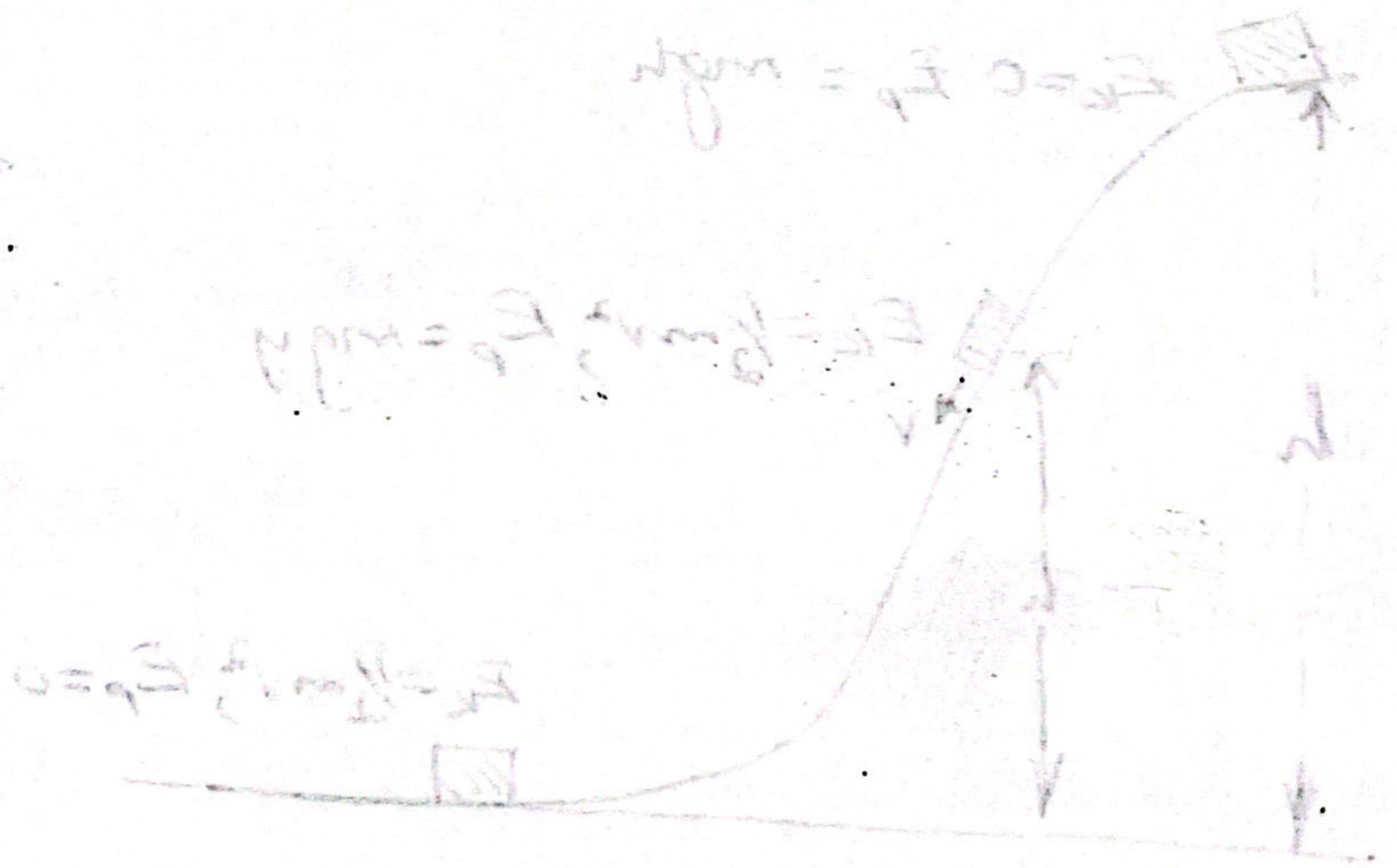
Chemical energy - obtained from food and fuels like coal, oil and gas.

Potential energy - is the energy of a body w.r.t position below it.

Kinetic energy is the energy due to motion.

Electrical energy - is due to flow of current.

Other forms of energy are heat, sound, light, magnetic and nuclear energy.

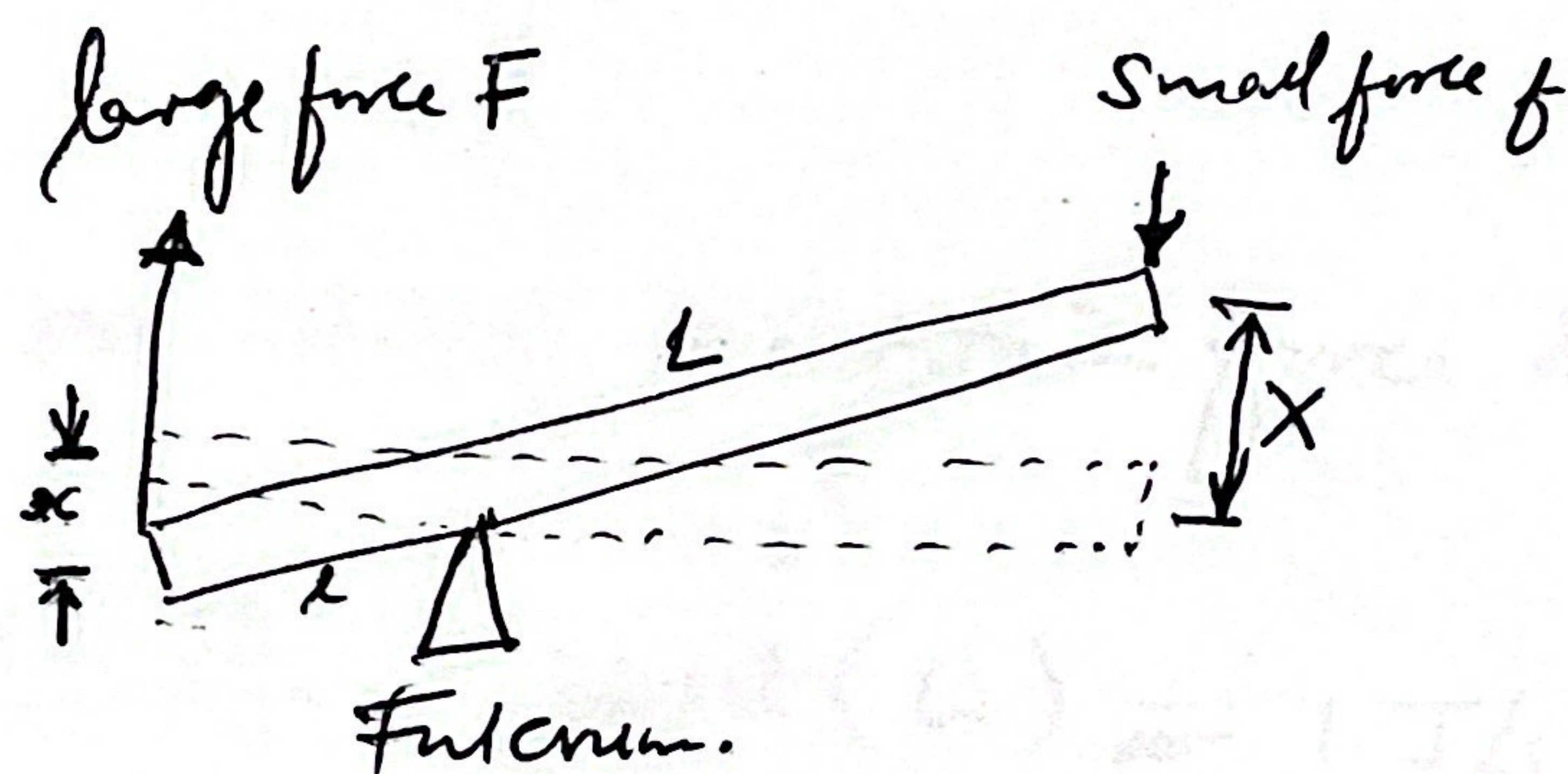


Electrostatic potential.

$$PE_E = k \frac{q_1 q_2}{r}$$

Labour saving devices.

The Lever



Work done is the same $= fX = Fx$

MA (mechanical Advantage) = ratio of output over input forces

$$MA = \frac{F}{f} = \frac{L}{l}$$

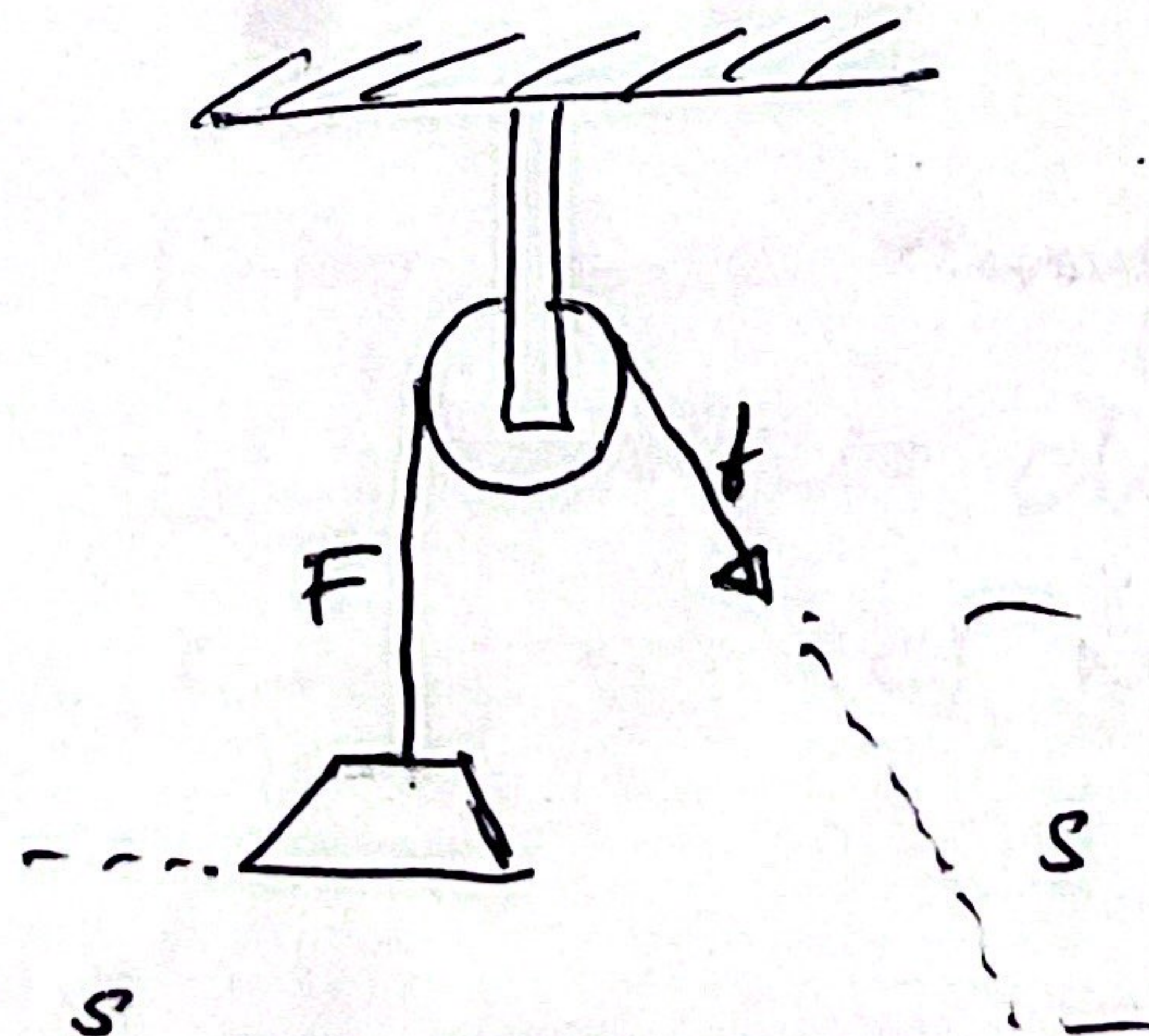
IF $f = 10\text{ N}$, $l = 0.5\text{ m}$, $L = 8\text{ m}$ then $MA = \frac{8}{0.5} = 16$ and
the force $F = 16 \times 10\text{ N} = 160\text{ N}$.

The pulley

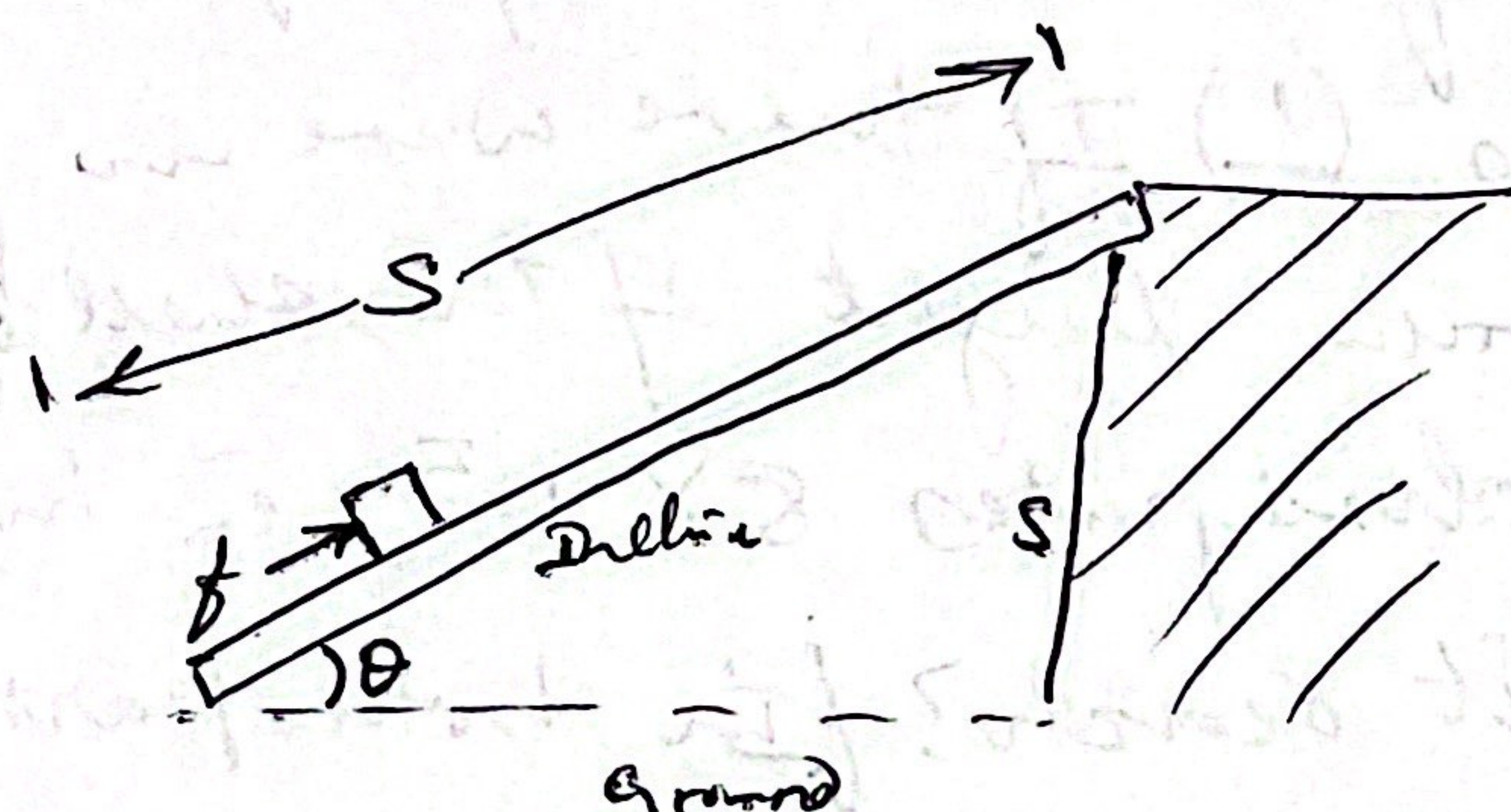
Simple case.

$$fS = Fs$$

$$MA = \frac{F}{f} = \frac{S}{s}$$



The incline



$$MA = \frac{F}{f} = \frac{s}{h} = \frac{1}{\sin \theta}$$

Power

power is the rate at which a certain force does work.

$$\boxed{P = \frac{W}{t}} \cdot \underline{\text{Units}} \quad \text{Watt (W)} = 1 \text{ J/s} = 1 \text{ Kg m}^2/\text{s}^3$$

When a force acts on a particle, the power expended on the particle is

$$P = \frac{\Delta W}{\Delta t} = F \cdot \frac{\Delta s}{\Delta t}$$

or

$$\boxed{P = F \cdot v}$$

Example.

As a girl lifts a 1-kg book through 1m, she does 12J of work on the book. What is its change of (a) potential energy U ? (b) kinetic energy K ? (c) mechanical energy, $K+U$?

A shell of mass 10 kg is fired vertically upward with an initial speed of 600 m/s . (1) If there were no friction, what would be the maximum height of the shell? (2) If on its way up the shell dissipates $8 \times 10^5\text{ J}$ in the form of heat, what height does it reach? (In both parts, ignore the variation of gravitational force with height).

Power is the rate at which work is done.

$$P = \frac{W}{t} = (v) \cdot \frac{W}{t} \quad \boxed{P = F \cdot v}$$

When a force acts on a particle, the power expended on the particle is

$$P = \frac{\Delta W}{\Delta t} = F \cdot \frac{\Delta x}{\Delta t}$$

$$\boxed{P = F \cdot v}$$

At a point, lift is a 2-kg object through 1 m, the work done is 20 J. What is the change of its potential energy? (a) 20 J (b) 40 J (c) 10 J (d) 5 J